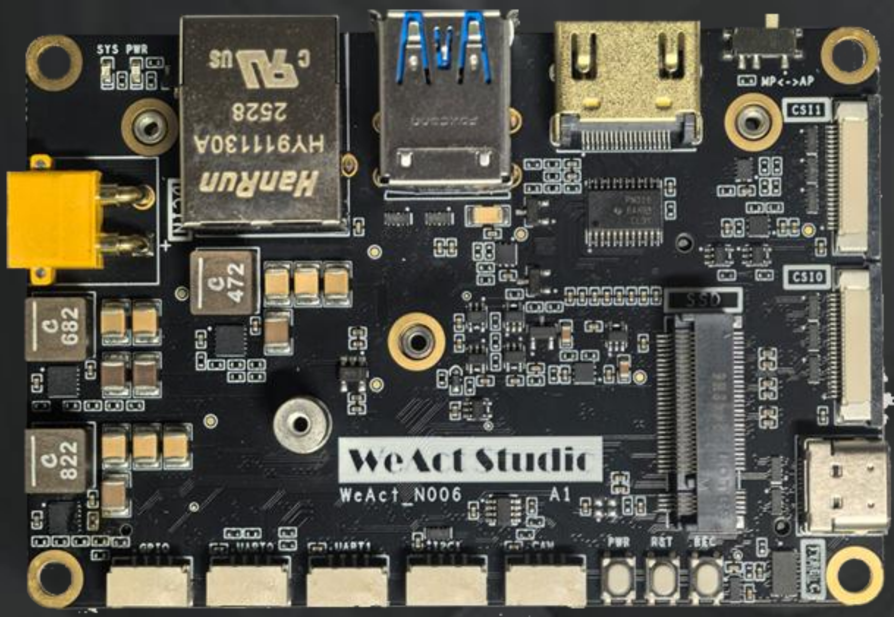


WeAct Studio

N006

载板/底板

使用教程

WeAct Studio

目录

REVISION HISTORY

Draft Date	Revision	Description
2026.02.24	V1.0	1. 初始版本
2026.06.07	V1.1	1. 支持 Jetpack7.x

1. 搭建烧写环境

a) 首先，需要一台装有 **Ubuntu20.04** 以上的电脑作为 HOST 端给 OrinNX 烧写系统，或者可以在 Windows 上安装 VMware 来实现。

*如果需要烧录 Jetpack 7.x，需要 Ubuntu24.04 电脑作为 HOST

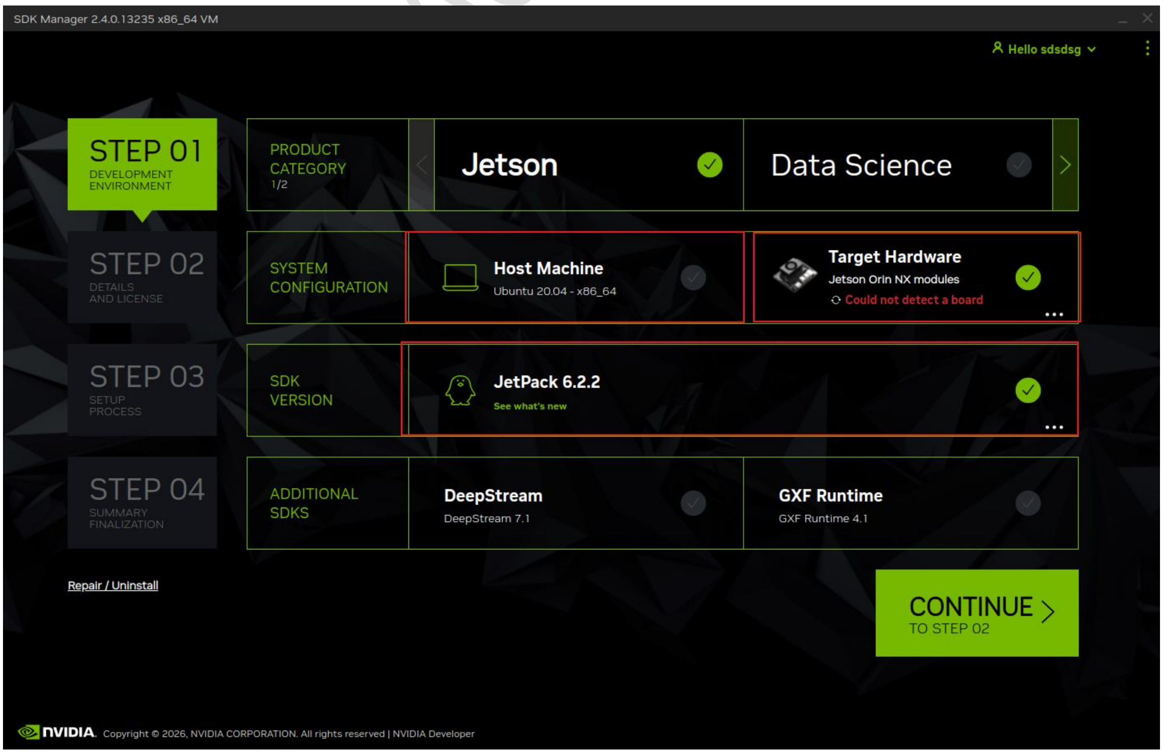
➢ VMware 上如何安装 Ubuntu20.04：
https://cloud.tencent.com/developer/article/2157601

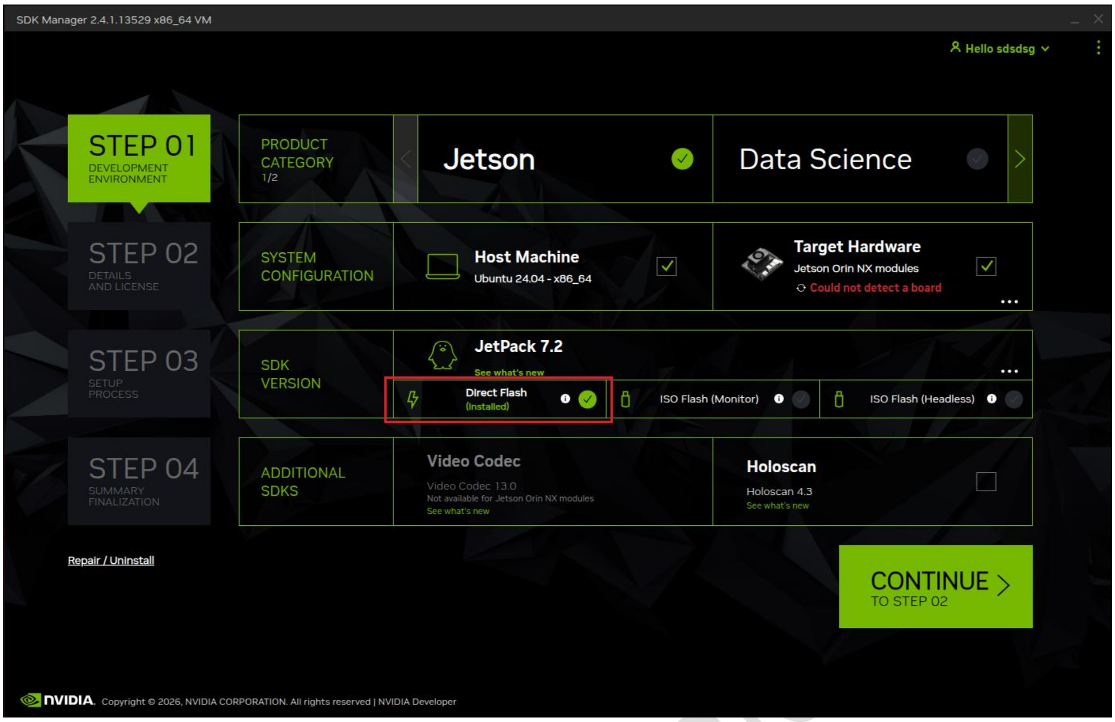
b) 在 NVIDIA 下载最新的 **SDK-Manager** 并在 ubuntu20.04 中安装（需要注册一个 NVIDIA 账号，后面也需要用到）

➢ SDK-Manager 下载地址：https://developer.nvidia.com/nvidia-sdk-manager

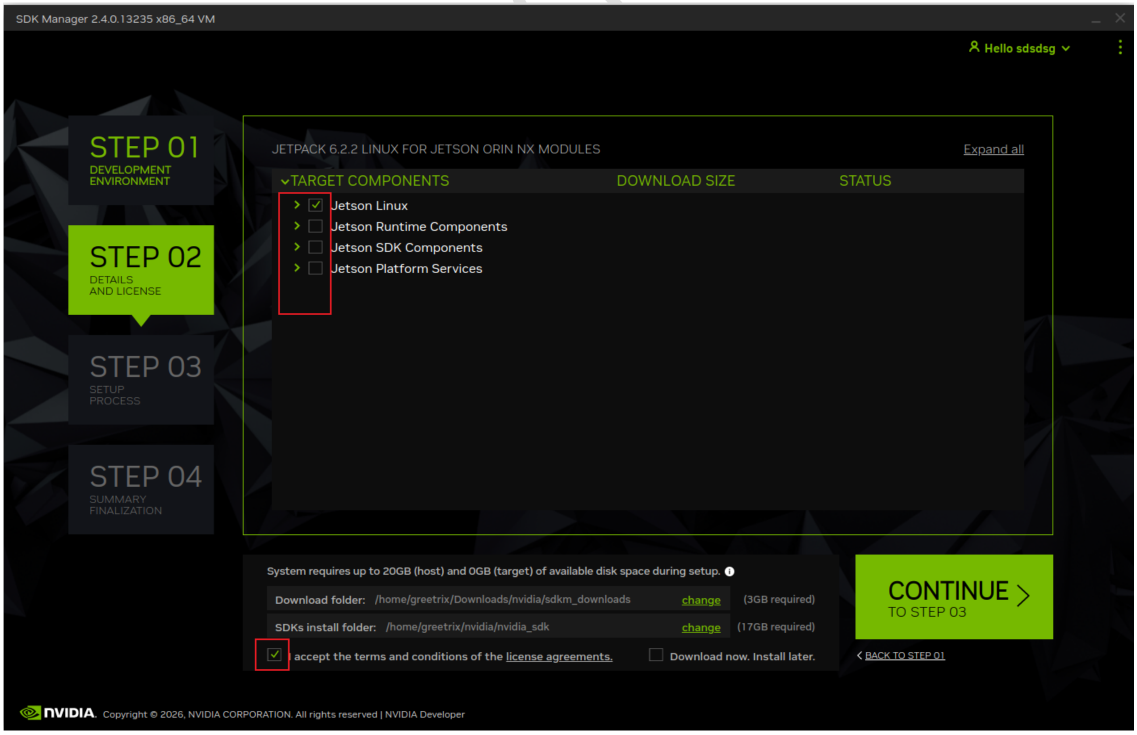
c) 选择需要 **Target Hardware** 以及 **JetPack** 版本，**不勾选 Host Machine（节省主机存储空间）**，DeepStream/GXF Runtime 根据组件需求勾选，这里以 **Jetpack6.2.2 版本为例**选择，点击 **Continue**

***对于 Jetpack 7.x，选择 Direct Flash**



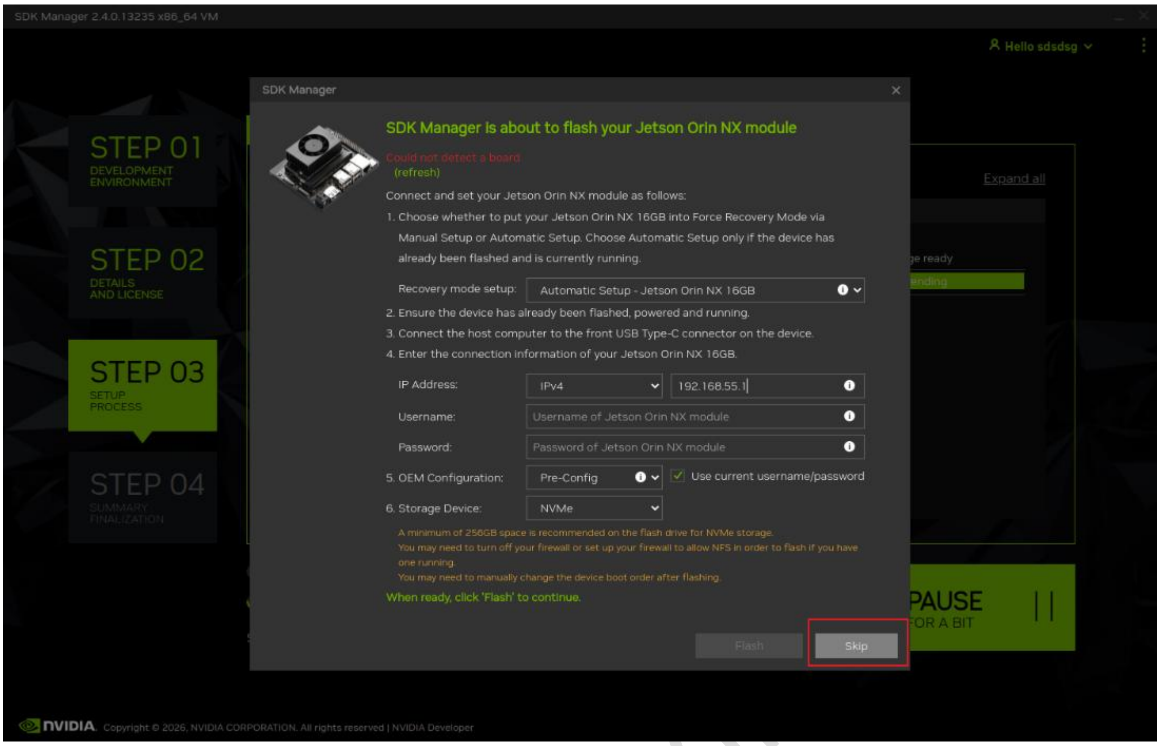


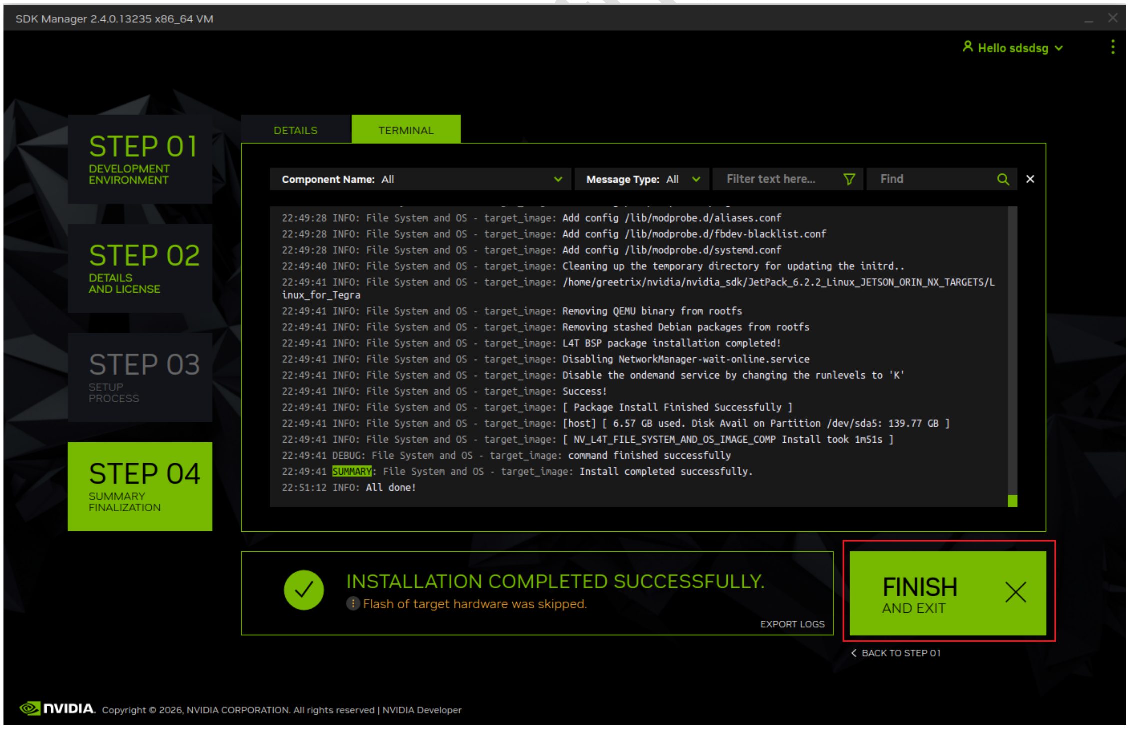
d) 这里只勾选 **Jetson Linux**，**取消勾选 Jetson Runtime/SDK Components/Platform Services（后面安装组件会重新下载并一起安装）**，点击 CONTINUE 进行下一步。



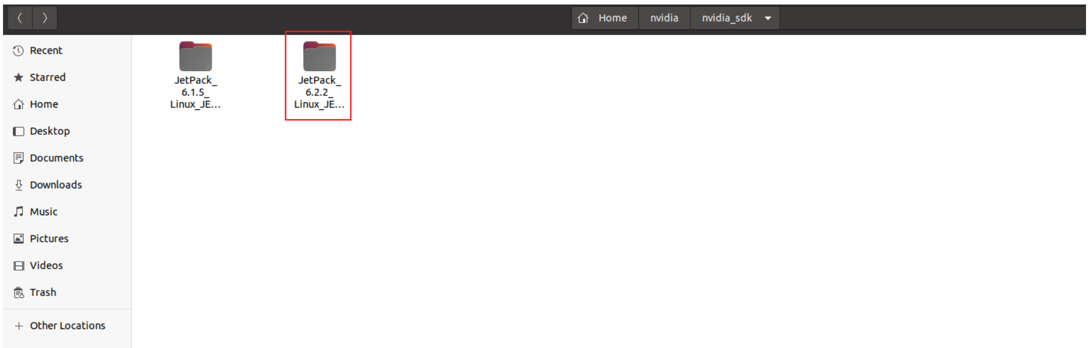
e) P.S：请在畅通的网络环境下进行下载以及安装，下载或安装失败时，可点击 **Retry** 继续，直至**全部状态为 Installed 并且显示绿色**，安装过程中会弹出联网烧写的信息，选择 **Skip。（后续步骤通过脚本来自动替换设备树并且烧写系统）**

WeAct Studio



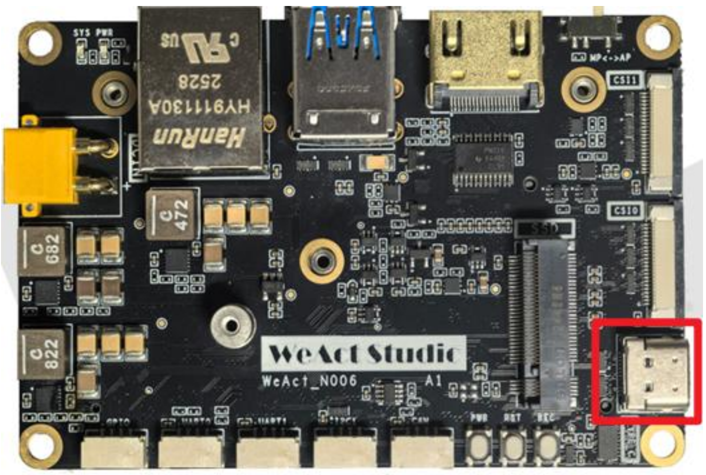


f) 安装成功后，会在**~/nvidia/nvidia_sdk/** 下有相应版本烧写所需的文件


Home
nvidia
nvidia_sdk
Recent
Starred
Home
Desktop
Documents
Downloads
Music
Pictures
Videos
Trash
Other Locations
JetPack_6.1.5_Linux_JE...
JetPack_6.2.2_Linux_JE...

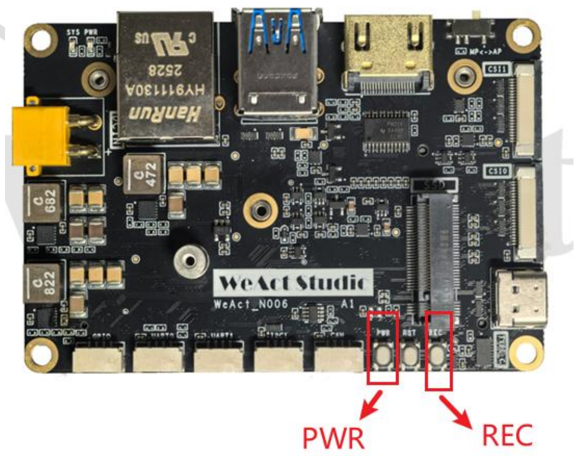
2. 核心板烧写系统

1. 使用 **USB Type-C** 线连接载板上的 **USB OTG** 接口



2. 将开机键**拨至 MP（手动开机）**，两种方式进入 Recovery 模式
   a) 按住 **REC** 键，再按一下 **PWR** 键开机，松开 **REC** 键进入 Recovery 模式
   b) 按一下 **PWR** 开机，按住 **REC** 键，按一下 **RST** 键进入 Recovery 模式

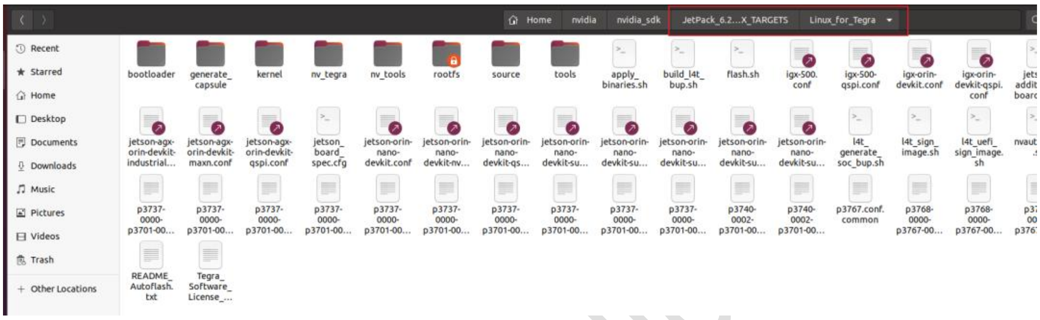
   此时 VMWare 右下角会出现 **NVIDIA 的 USB 驱动标志**，或者打开终端，输入 **lsusb** 命令，会发现 **Nvidia Corp**，**同时风扇转速会到达最大。**




```
greetrix@ubuntu:~/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra$ lsusb
Bus 004 Device 001: ID 1d6b:0003 Linux Foundation 3.0 root hub
Bus 003 Device 004: ID 0e0f:0002 VMware, Inc. Virtual USB Hub
Bus 003 Device 003: ID 0e0f:0002 VMware, Inc. Virtual USB Hub
Bus 003 Device 005: ID 0955:7323 NVIDIA Corp. APX
Bus 003 Device 002: ID 0e0f:0003 VMware, Inc. Virtual Mouse
Bus 003 Device 001: ID 1d6b:0002 Linux Foundation 2.0 root hub
Bus 001 Device 001: ID 1d6b:0002 Linux Foundation 2.0 root hub
Bus 002 Device 003: ID 0e0f:0002 VMware, Inc. Virtual USB Hub
Bus 002 Device 002: ID 0e0f:0008 VMware, Inc. Virtual Bluetooth Adapter
Bus 002 Device 001: ID 1d6b:0001 Linux Foundation 1.1 root hub
greetrix@ubuntu:~/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra$
```

*VMwave 会有对应的图标显示

3. 进入**~/nvidia/nvidia_sdk/JetPack_<Version>_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra**，<Version>为 Jetpack 版本号，此处以 6.2.2 为例，打开终端：

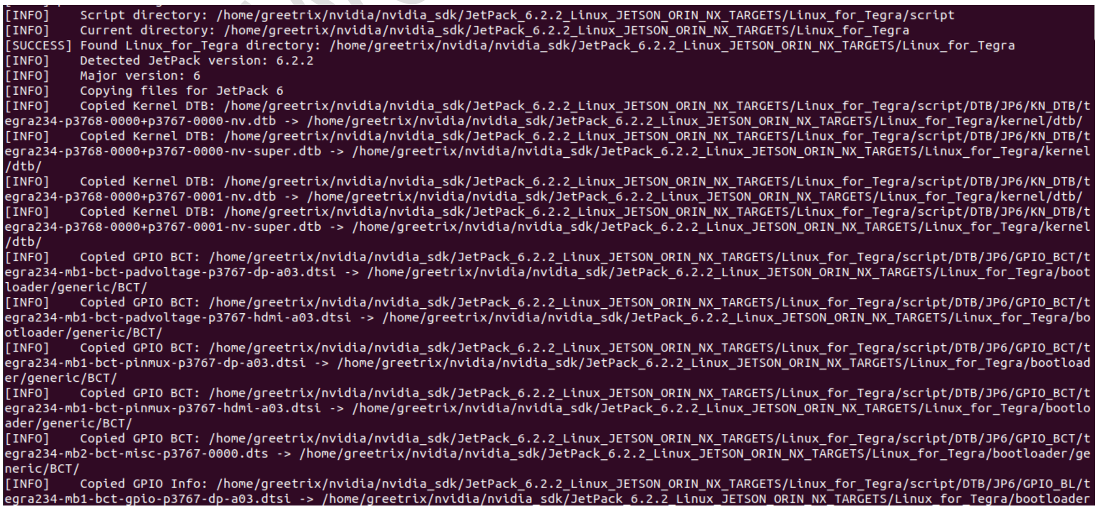


a) sudo apt-get install git
b) git clone https://github.com/WeActStudio/WeAct_N006
c) sudo bash WeAct_N006/03_Script/weact_n006_orin_flash.sh

或

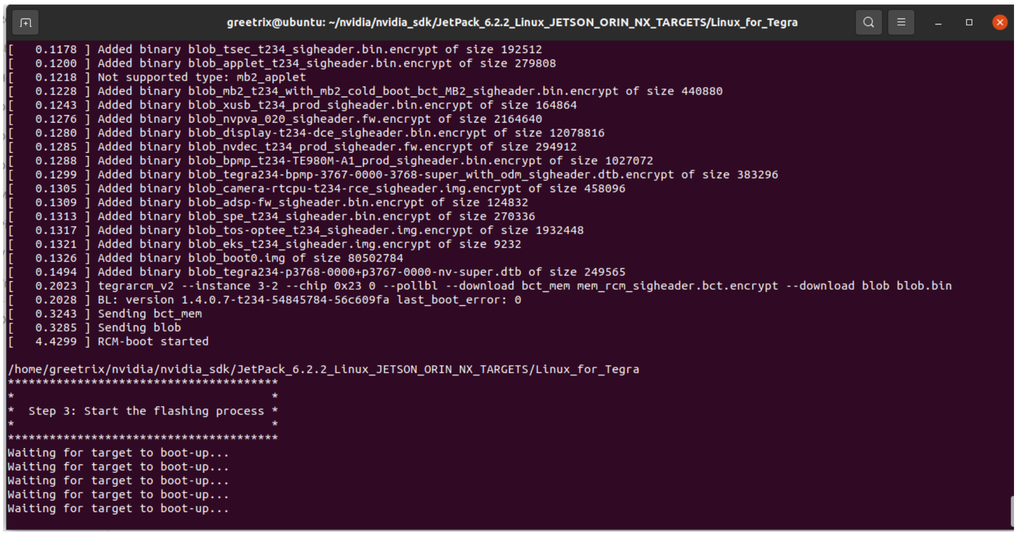
b) git clone https://gitee.com/WeAct-TC/weact_n006

c) sudo bash weact_n006/03_Script/weact_n006_orin_flash.sh

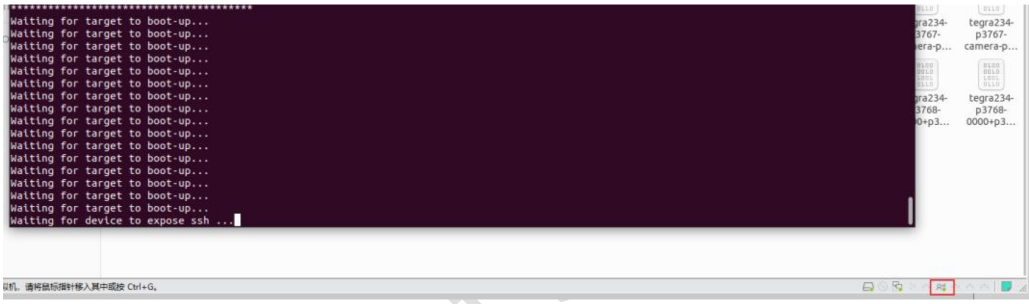


```
[INFO]    Script directory: /home/greetrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/script
[INFO]    Current directory: /home/greetrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra
[SUCCESS] Found Linux_for_Tegra directory: /home/greetrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra
[INFO]    Detected JetPack version: 6.2.2
[INFO]    Major version: 6
[INFO]    Copying files for JetPack 6
[INFO]    Copied Kernel DTB: /home/greetrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/script/DTB/JP6/KN_DTB/tegra234-p3768-0000+p3767-0000-nv.dtb -> /home/greetrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/kernel/dtb/
[INFO]    Copied Kernel DTB: /home/greetrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/script/DTB/JP6/KN_DTB/tegra234-p3768-0000+p3767-0000-nv-super.dtb -> /home/greetrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/kernel/dtb/
[INFO]    Copied Kernel DTB: /home/greetrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/script/DTB/JP6/KN_DTB/tegra234-p3768-0000+p3767-0001-nv.dtb -> /home/greetrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/kernel/dtb/
[INFO]    Copied Kernel DTB: /home/greetrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/script/DTB/JP6/KN_DTB/tegra234-p3768-0000+p3767-0001-nv-super.dtb -> /home/greetrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/kernel/dtb/
[INFO]    Copied GPIO BCT: /home/greetrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/script/DTB/JP6/GPIO_BCT/tegra234-mb1-bct-padvoltage-p3767-dp-a03.dtsi -> /home/greetrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/bootloader/generic/BCT/
[INFO]    Copied GPIO BCT: /home/greetrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/script/DTB/JP6/GPIO_BCT/tegra234-mb1-bct-padvoltage-p3767-hdmi-a03.dtsi -> /home/greetrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/bootloader/generic/BCT/
[INFO]    Copied GPIO BCT: /home/greetrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/script/DTB/JP6/GPIO_BCT/tegra234-mb1-bct-pinmux-p3767-dp-a03.dtsi -> /home/greetrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/bootloader/generic/BCT/
[INFO]    Copied GPIO BCT: /home/greetrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/script/DTB/JP6/GPIO_BCT/tegra234-mb1-bct-pinmux-p3767-hdmi-a03.dtsi -> /home/greetrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/bootloader/generic/BCT/
[INFO]    Copied GPIO BCT: /home/greetrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/script/DTB/JP6/GPIO_BCT/tegra234-mb2-bct-misc-p3767-0000.dts -> /home/greetrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/bootloader/generic/BCT/
[INFO]    Copied GPIO Info: /home/greetrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/script/DTB/JP6/GPIO_BL/tegra234-mb1-bct-gpio-p3767-dp-a03.dtsi -> /home/greetrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/bootloader
```

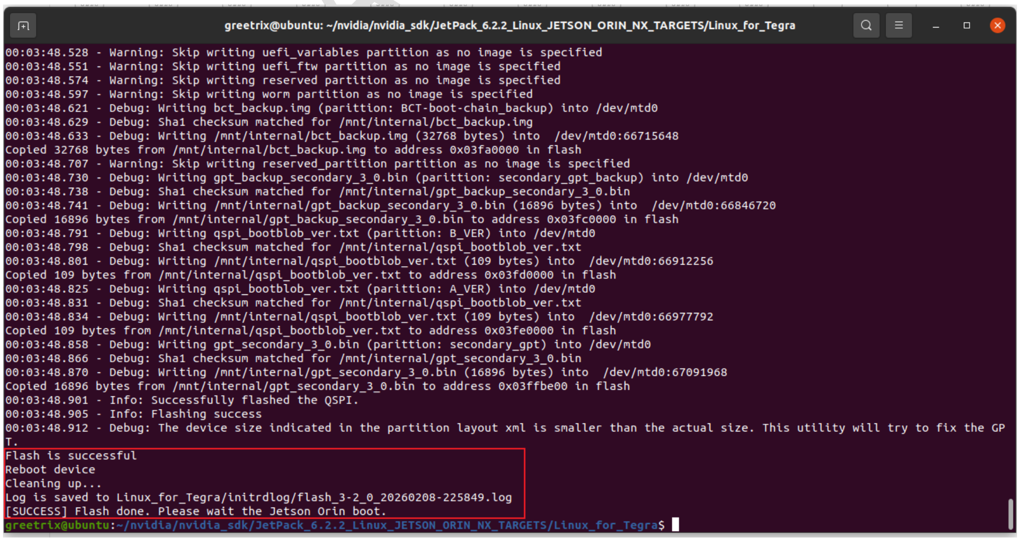
脚本会自动替换设备树并且烧写已开启 Super Mode 的系统



此处 USB 会切换，如果是虚拟机，请提前给 USB 自动连接的权限

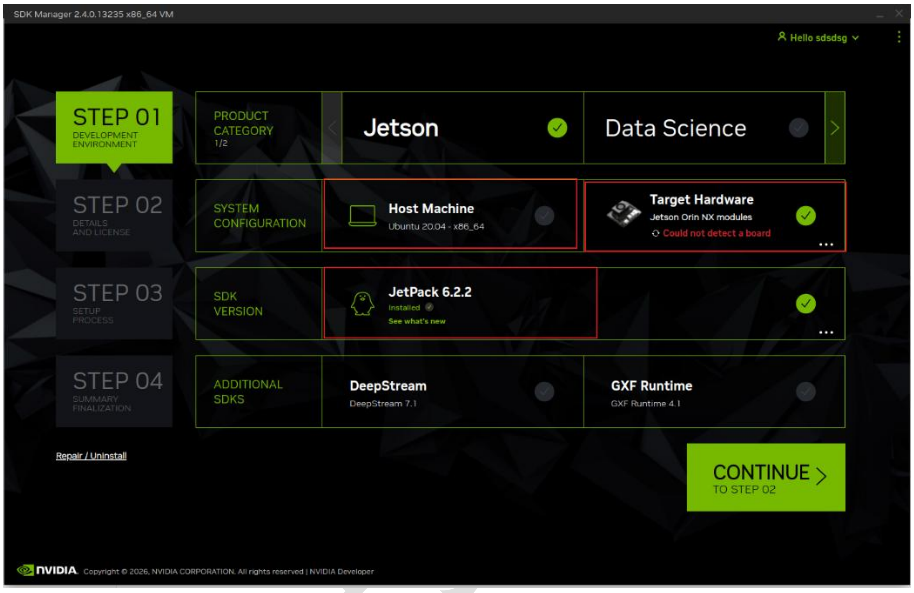


烧写系统成功后会有 Successfully!显示，如下图所示。

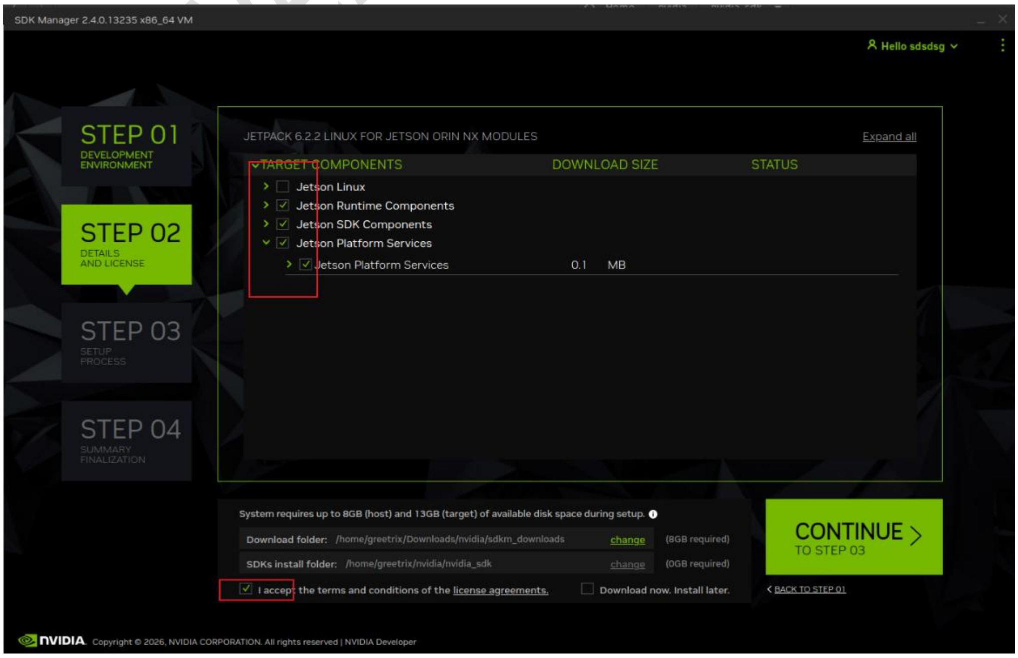


3.　安装 NVIDIA 组件

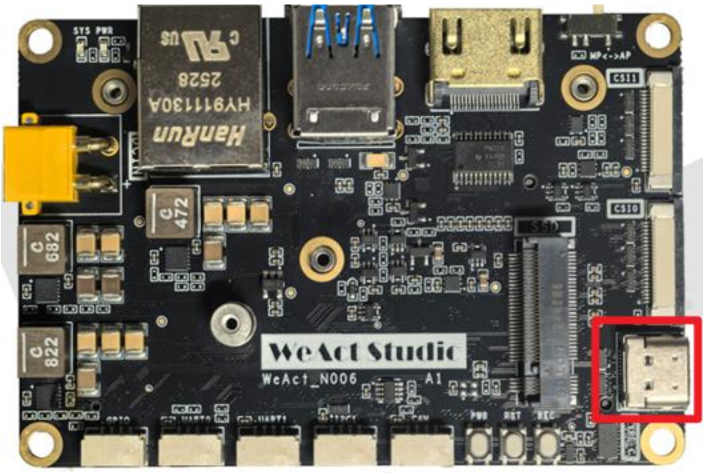
1.　选择需要 **Target Hardware** 以及 **JetPack** 版本，**不勾选 HostMachine**，，点击 **Continue**



2.　勾选所需要 **SDK 组件**，勾选 **I accept the terms and conditions of the license agreements**，点击 **CONTINUE** 进行下一步。

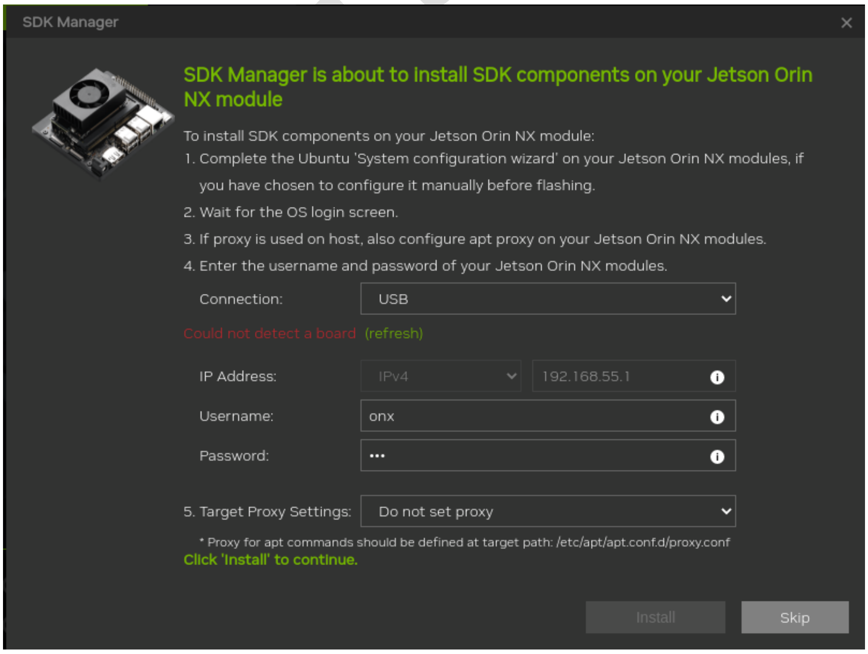


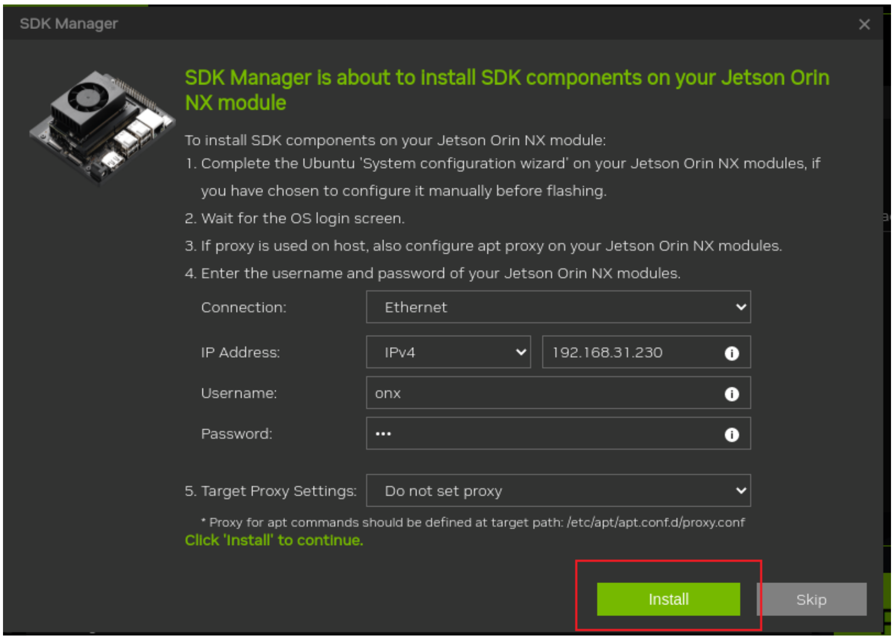
3.　　使用 **USB Type-C** 线连接载板上的 **USB OTG** 接口 。



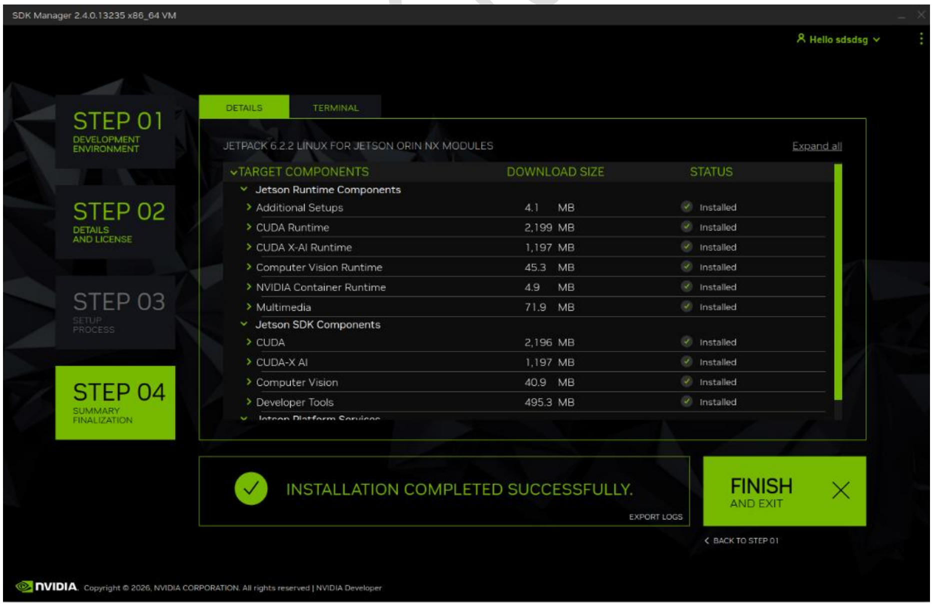
4.　　将开机键**拨至 MP（手动开机）**，按一下 **PWR** 键开机，此时 VMWare 右下角会出现 **NVIDIA 的 USB 驱动标志**，或者打开终端，输入 **lsusb** 命令，会发现 **Nvidia Corp**。

5.　　输入 **Orin NX** 账号密码，**Orin NX** 端请**保持联网状态，也可以用同局域网下的 IP（图 2，主机和 ORIN NX 连接同一个路由器）**




SDK Manager
SDK Manager is about to install SDK components on your Jetson Orin NX module
To install SDK components on your Jetson Orin NX module:
1. Complete the Ubuntu 'System configuration wizard' on your Jetson Orin NX modules, if you have chosen to configure it manually before flashing.
2. Wait for the OS login screen.
3. If proxy is used on host, also configure apt proxy on your Jetson Orin NX modules.
4. Enter the username and password of your Jetson Orin NX modules.
Connection:
Ethernet
IP Address:
IPv4
192.168.31.230
Username:
onx
Password:
5. Target Proxy Settings:
Do not set proxy
* Proxy for apt commands should be defined at target path: /etc/apt/apt.conf.d/proxy.conf
Click 'Install' to continue.
Install
Skip

6. 等待安装完成即可。


SDK Manager 2.4.0.13235 x86_64 VM
Hello sdsdsg
STEP 01
DEVELOPMENT ENVIRONMENT
STEP 02
DETAILS AND LICENSE
STEP 03
SETUP PROCESS
STEP 04
SUMMARY FINALIZATION
DETAILS
TERMINAL
JETPACK 6.2.2 LINUX FOR JETSON ORIN NX MODULES
Expand all
TARGET COMPONENTS
DOWNLOAD SIZE
STATUS
Jetson Runtime Components
Additional Setups 4.1 MB Installed
CUDA Runtime 2,199 MB Installed
CUDA X-AI Runtime 1,197 MB Installed
Computer Vision Runtime 45.3 MB Installed
NVIDIA Container Runtime 4.9 MB Installed
Multimedia 71.9 MB Installed
Jetson SDK Components
CUDA 2,196 MB Installed
CUDA-X AI 1,197 MB Installed
Computer Vision 40.9 MB Installed
Developer Tools 495.3 MB Installed
INSTALLATION COMPLETED SUCCESSFULLY.
EXPORT LOGS
FINISH
AND EXIT
BACK TO STEP 01
NVIDIA
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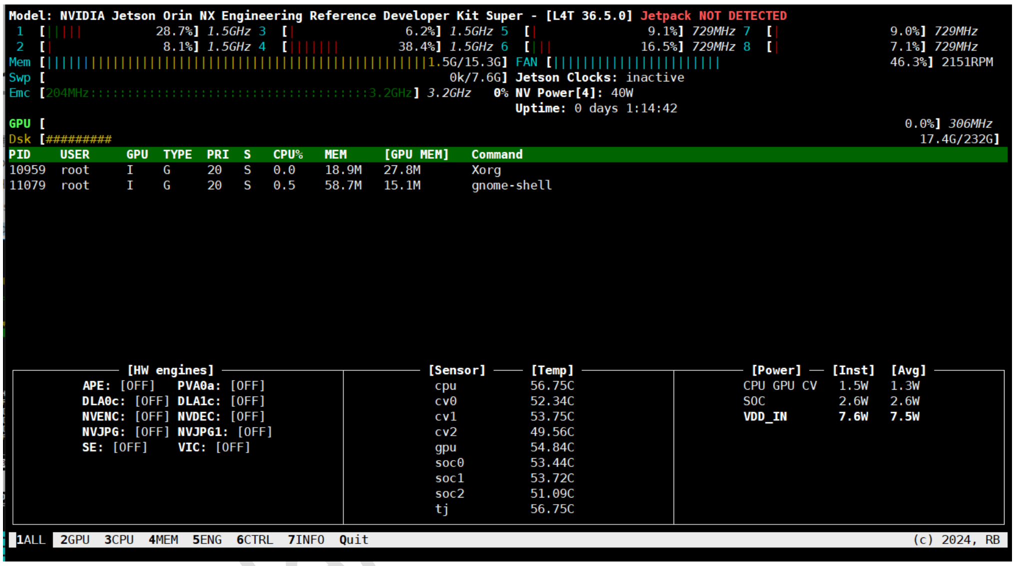
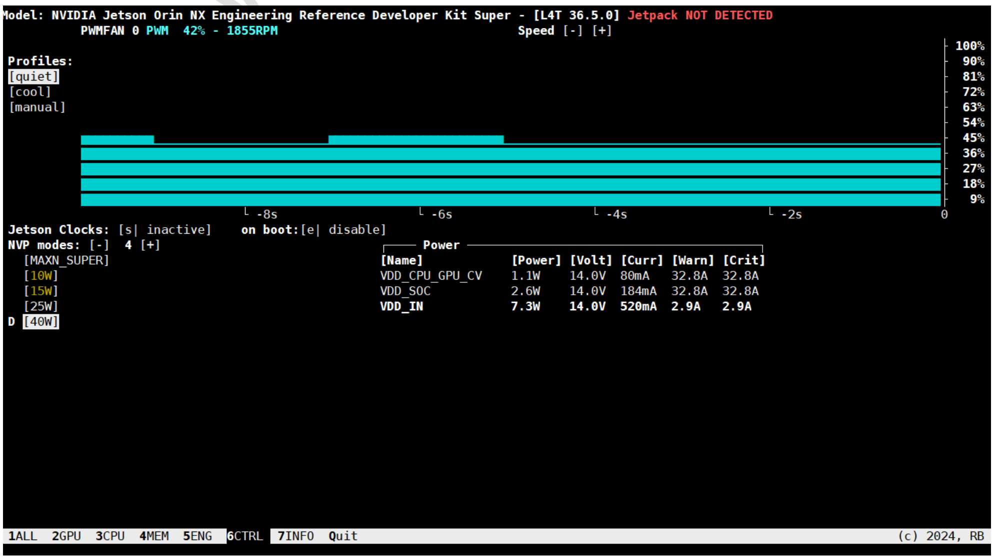
4. 安装 JTOP（控制风扇，查看系统信息）

1. 打开终端，通过下面命令安装 JTOP
   a) sudo apt-get install python3-pip python3-dev -y
   b) sudo -H pip3 install jetson-stats --break-system-packages
   c) sudo systemctl restart jtop.service
   d) sudo jtop



2. 点击 6CTRL，即可手动控制风扇转速及电源方案



5. 使用 CAN 进行通信

1. OrinNX 上集成了 1 个 CAN 控制器 CAN0，另外 WeAct Studio 的载板上设计了 1 个 CAN 收发器（CAN0），可**直接挂载 CAN 物理总线**使用。

2. OrinNX 自带 canbus 的驱动并集成到了镜像中，已经支持 canbus 无需多做处理。我们需要安装 canbus 模块。（在终端输入下面命令或者放入 rc.local 里面开启自启）

```
sudo modprobe can
sudo modprobe can-raw
sudo modprobe can-bcm
sudo modprobe can-gw
sudo modprobe can_dev
sudo modprobe mttcan
```


3. 通过 **lsmod** 检查是否安装成功。

```
root@onx:/home/onx# lsmod | grep can
can_gw                 24576  0
can_bcm                32768  0
can_raw                28672  0
can                    28672  3 can_gw,can_raw,can_bcm
mttcan                 65536  0
nvpps                  28672  1 mttcan
can_dev                36864  1 mttcan
root@onx:/home/onx#
```

4. 配置 canbus 属性，和串口的波特率设置类似。

```
sudo ip link set can0 type can bitrate 500000
sudo ip link set up can0
```

5. 通过 ifconfig 查看是否配置成功。

```
root@onx:/home/onx# sudo ip link set can0 type can bitrate 500000
root@onx:/home/onx# sudo ip link set up can0
root@onx:/home/onx# ifconfig
can0: flags=193<UP,RUNNING,NOARP>  mtu 16
        unspec 00-00-00-00-00-00-00-00-00-00-00-00-00-00-00-00  txqueuelen 10  (UNSPEC)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
        device interrupt 201
```

6. 在一个终端通过 cansend can0(can1) ×××命令来发送数据, 另一个终端通过 candump can1(can0)完成实际信号收发测试

```
root@onx:/home/onx# candump can0
  can0  555   [2]  AA AA
  can0  555   [2]  AA AA
  can0  555   [2]  AA AA
  can0  555   [2]  AA AA
  can0  555   [2]  AA AA
  can0  555   [2]  AA AA
```

```
root@onx:/home/onx# cansend can0 555#AAAA
root@onx:/home/onx# cansend can0 555#AAAA
root@onx:/home/onx# cansend can0 555#AAAA
root@onx:/home/onx# cansend can0 555#AAAA
root@onx:/home/onx# cansend can0 555#AAAA
root@onx:/home/onx# cansend can0 555#AAAA
root@onx:/home/onx#
```

6. GPIO 在 SHELL 中的使用

1. Jetson OrinNX 可直接通过 shell 命令控制 GPIO 输入输出，通过下面命令安装 gpiod
   a) sudo apt-get install gpiod
   b) sudo apt-get install libgpiod-dev

2. GPIO 使用方法

	Orin NX 8G/16G
GPIO1	PG.06
GPIO2	PAC.06
GPIO3	PN.01
GPIO4	PH.00

   a. GPIO 输出：以 GPIO1 为例
      i. gpioset `gpiofind "PG.06"`=0 （设置输出低电平）
      ii. gpioset `gpiofind "PG.06"`=1 （设置输出高电平）
   b. GPIO 输入：以 GPIO1 为例
      i. gpioget `gpiofind "PG.06"`

7. UART/I2C/CSI 的使用

1. UART 的使用

	Jetpack 5.X	Jetpack 6.X/7.X
UART0	ttyTHS1	ttyTHS0
UART1	ttyTHS0	ttyTHS1

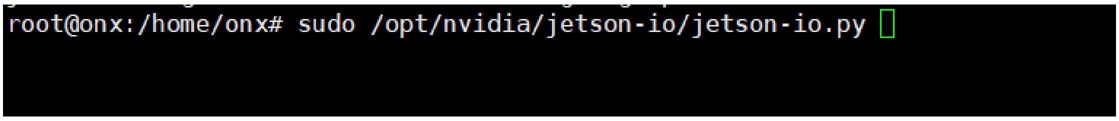
a. 可以通过下面命令安装 cutecom/minicom 来使用 UART:
   a) sudo apt-get install cutecom
   b) sudo apt-get install minicom

2. **I2C 的使用** (载板上 I2C 对应为 I2C@7)，可以通过下面命令来查看挂在在 I2C 接口上的设备地址:

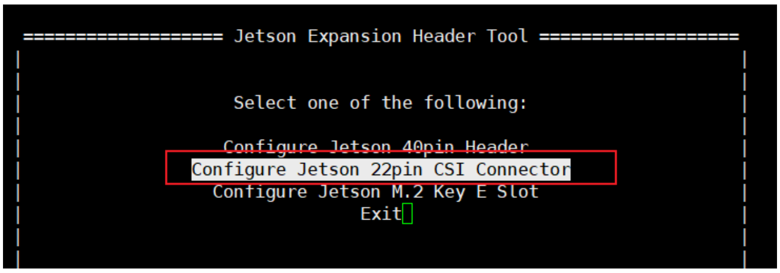
a) sudo i2cdetect -r -y 7

3. **CSI 的使用**（以双 IMX219 为例）

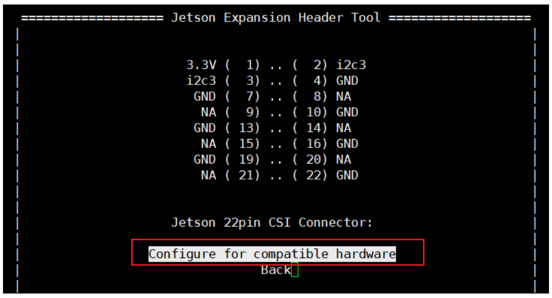
a) 命令行中输入: **sudo /opt/nvidia/jetson-io/jetson-io.py**



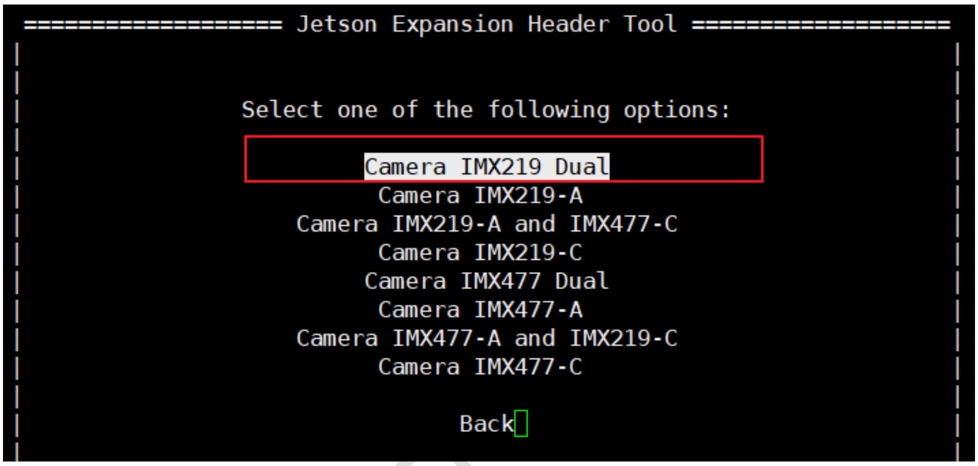
b) 选择 **Configure Jetson 22pin CSI Connector**



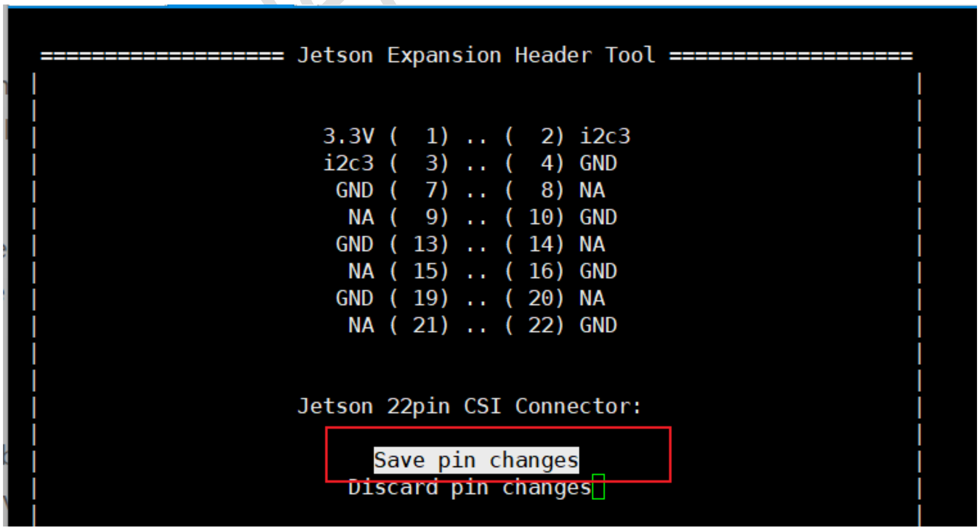
c) 选择 **Configure for compatible hardware**



d) 选择 **Camera IMX219 Dual**



e) 选择 **Save pin changes**



f) 选择 **Save and reboot to reconfigure pins**

```
=================== Jetson Expansion Header Tool ===================
|                                                                  |
|                                                                  |
|                  Select one of the following:                    |
|                                                                  |
|                 Configure Jetson 40pin Header                    |
|            Re-configure Jetson 22pin CSI Connector               |
|                Configure Jetson M.2 Key E Slot                   |
|              Save and reboot to reconfigure pins                 |
|                Save and exit without rebooting                  |
|                    Discard all pin changes                       |
|                             Exit                                 |
|                                                                  |
```

g） 等待重启即可，即可使用双 IMX219

联系我们

1. Github：https://github.com/WeActStudio
2. 码云：https://gitee.com/WeAct-TC
3. 淘宝：https://weactstudio.taobao.com/

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